

Shayan Amir Shahkarami

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PROFILE

DevOps Engineer and Backend Developer with hands-on experience in container infrastructure, Kubernetes orchestration, and cloud-based backend services. Focused on building scalable systems using Go and Python, with experience in AWS and containerized architectures. Strong interest in Machine Learning and Large Language Models. Currently pursuing a Master's degree in Computer Science at Heidelberg University.

WORK EXPERIENCE

DevOps Engineer May 2025 – Apr. 2026
Container Infrastructure & Security Iran

- Designed and maintained Docker images for multiple production services with a focus on security hardening and reproducibility
- Deployed and operated containerized applications using Kubernetes (Deployments, Services, ConfigMaps, Secrets)
- Took full responsibility for container and cluster security, including vulnerability scanning and runtime protection

Security Researcher & Developer Jul. 2022 – Oct. 2024
Backend Development & Security Tools Iran

- Developed and containerized custom security tools for vulnerability assessment and penetration testing
- Implemented RESTful APIs and backend services in Go following clean architecture principles
- Created extensive test cases for backend services and APIs in Go to ensure correctness and reliability

Internship – IT Helpdesk & User Support Nov. 2022 – Jan. 2023
IT Infrastructure & Support Iran

- Provided first-level IT support and troubleshooting for end users
- Assisted with network infrastructure tasks including switch configuration and network setup

Internship – Cybersecurity Tools & Attack Methods Feb. 2023 – May 2023
Cybersecurity & Penetration Testing Iran

- Worked hands-on with industry-standard cybersecurity tools for threat analysis and penetration testing
- Applied fundamental concepts of cybersecurity, attack vectors, and system defense

PROJECTS

Alferd – Personal AI Assistant | Go, Python, LLMs, Docker 2024 – Present

- Developed a personal AI assistant focused on local execution and efficiency, integrating LLMs into a modular, containerized architecture
- Designed a plugin-based system in Go for extensibility along with Python components for ML inference

SmartFuzzer – Intelligent Security Fuzzer | Python, Machine Learning 2023 – 2024

- Developed an ML-powered fuzzer for automated vulnerability discovery in application input fields
- Applied supervised learning techniques to prioritize high-risk inputs and improve test coverage compared to conventional fuzzing methods

EDUCATION

Heidelberg University Apr. 2026 – Present
M.Sc. Data & Computer Science Heidelberg, Germany

Islamic Azad University, Central Tehran Branch Oct. 2020 – Oct. 2024
Bachelor of Engineering – Computer Science Tehran, Iran

CERTIFICATIONS & TRAINING

CompTIA Security+ – CanDo Academy, Jun. 2022

Credential ID: 01/43154

CCNA 200-301 – CanDo Academy, Mar. 2023

Credential ID: 01/51339

Certified Ethical Hacker (CEH) – Nooranet Academy, Nov. 2022

CID: 20221207-54-SK6436

Certified Security Professional (CSP) – Nooranet Academy, Nov. 2022

CID: 20221208-69-SG6781

Red Team Expert – Nooranet Academy, Feb. 2023

CID: 20230413-22-XD5154

SKILLS

Programming Languages : Go, Python, C#, C++, Kotlin

DevOps & Cloud : Docker, Kubernetes, GitLab CI/CD, AWS

AI & Machine Learning : PyTorch, TensorFlow, FastAI, LLMs, Transformers, FastAPI

Databases : PostgreSQL, MySQL

Languages : German (B2 – Goethe-Institut, 71/100), English (TOEFL iBT 100), Persian (Native)